rectangle, often called a *georectangle*, by taking the coordinates of the top-left and bottom-right corners. These coordinates consist of a pair of longitude and latitude measurements representing the corners.

Elasticsearch provides a `bounding_box` query that lets us search for required addresses in a georectangle. This query fetches the points of interest (as query criteria) inside the georectangle constructed by our set of coordinates. For example, figure 12.1 shows the addresses in central London enclosed in one such georectangle. Addresses intersecting this rectangle are returned as positive results. We run through some `bounding_box` queries in detail shortly.

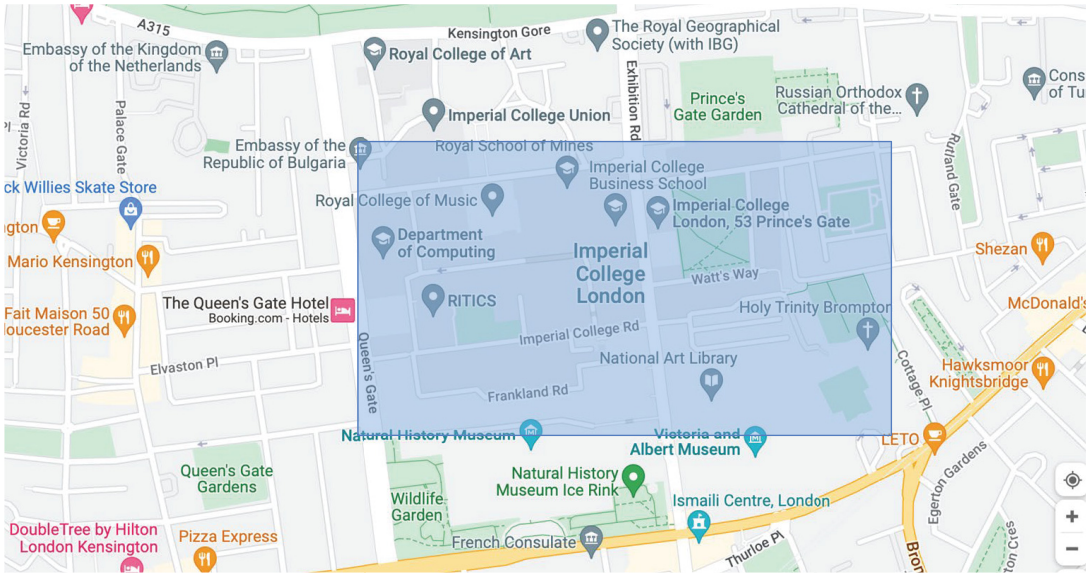


Figure 12.1 A georectangle constructed with a set of longitude and latitude coordinates

12.1.2 The geo_distance query

You may have seen Hollywood movies where an FBI agent tries to pin down a fugitive in an area drawn as a circle around a central focal point. That's what the `geo_distance` query does!

Elasticsearch provides the `geo_distance` query to fetch addresses in an area enclosed by a circle. The center is defined by the longitude, the latitude, and a radius as the distance. In figure 12.2, we have a central location (shown as the dropped pin on the map) and a circular area covering the addresses we are looking for. The focus (or central location) is a point on the map dictated by latitude and longitude coordinates.